PROJECT SYNOPSIS REPORT

Title: Loan Approval Prediction Using Machine Learning

1. Introduction

Loan approval is a high-impact decision process in banking and finance. Manual screening can be slow and inconsistent when many applications need to be processed quickly. This project builds a machine learning based prediction system that estimates whether a loan should be approved based on applicant profile and financial indicators. The system demonstrates an end-to-end workflow from data preprocessing to model deployment through a web interface.

2. Problem Statement

Financial institutions receive large numbers of loan applications with mixed data quality, missing values, and diverse applicant backgrounds. The objective is to develop a reliable binary classification model that predicts Loan_Status (Approved/Rejected) from applicant attributes, improves consistency of decisions, and supports faster preliminary screening.

3. Objectives

- Build a robust supervised ML pipeline for loan approval prediction.
- Handle missing values and categorical variables systematically.
- Evaluate model performance using standard classification metrics.
- Save a reusable trained model pipeline for future predictions.
- Provide an interactive Streamlit web application for real-time inference.

4. Dataset Description

The project uses a CSV dataset (data/loan_data.csv) with 11 input features and 1 target column. Target column Loan_Status is normalized to binary form where Y/1 -> 1 (Approved) and N/0 -> 0 (Rejected).

Feature columns used:

- Gender

- Married

- Dependents

- Education

- Self_Employed

- ApplicantIncome

- CoapplicantIncome

- LoanAmount

- Loan_Amount_Term

- Credit_History

- Property_Area

5. Methodology and System Design

5.1 Data Validation and Target Normalization

The training script verifies required columns before modeling. Unsupported target values raise explicit errors, which improves data integrity and reproducibility.

5.2 Preprocessing Pipeline

Preprocessing is implemented with sklearn ColumnTransformer and Pipeline:

- Categorical columns: SimpleImputer(strategy='most_frequent') followed by OneHotEncoder(handle_unknown='ignore').

- Numerical columns: SimpleImputer(strategy='median').

This design prevents data leakage by fitting transformations only on training data and applying the same transformations during inference.

5.3 Model Selection

Classifier: RandomForestClassifier with n_estimators=200, max_depth=10, random_state=42, class_weight='balanced'.

Random Forest is selected for its strong non-linear learning ability, robustness to mixed feature types, and good practical performance for tabular classification.

5.4 Training and Evaluation Strategy

- Train-test split: 80:20 with random_state=42 and stratify=y to preserve class proportion.

- Training shape: 392 rows x 11 features

- Test shape: 99 rows x 11 features

- Evaluation metrics: Accuracy, Precision, Recall, F1-score, Confusion Matrix, Classification Report.

6. Performance Results

- Accuracy: 77.78%

- Precision: 82.19%

- Recall: 86.96%

- F1-score: 84.51%

- Confusion Matrix: [[17, 13], [9, 60]]

Interpretation:

The model achieves solid overall performance with stronger detection of approved loans than rejected loans. Precision and recall for class 1 (Approved) are comparatively high, while class 0 performance is lower, indicating scope to improve minority-class discrimination through tuning or rebalancing.

7. Deployment and User Interface

The project includes a Streamlit frontend (app.py) where users enter applicant details through form inputs. On submit, the app loads the saved model pipeline and returns:

- Prediction label (Approved/Rejected)

- Prediction value (1/0)

- Approval probability (if supported by model)

This makes the system practical for demonstration and real-time testing.

8. Project Structure

- src/train.py: data loading, preprocessing, model training, and metric export
- src/predict.py: reusable single-record prediction helper
- app.py: Streamlit web app for inference
- models/loan_pipeline.joblib: trained end-to-end pipeline artifact
- models/metrics.json: saved evaluation results
- data/loan_data.csv: dataset

9. Key Strengths

- End-to-end ML lifecycle in a clean modular structure.
- Leakage-safe preprocessing through integrated pipeline design.
- Serialized model for repeatable inference without retraining.
- Practical web deployment for project demonstration.

10. Limitations

- Performance for rejection class is weaker than approval class.
- No systematic hyperparameter optimization yet.
- Explainability is limited; feature importance reporting is not included in the app.
- Dataset quality and representativeness may constrain generalization.

11. Future Scope

- Hyperparameter tuning with GridSearchCV/RandomizedSearchCV.
- Compare additional algorithms (XGBoost, LightGBM, CatBoost).

- Add explainability tools (SHAP/feature importance dashboard).

- Introduce model monitoring and periodic retraining.

- Extend app with authentication and prediction-history database.

12. Conclusion

This project successfully implements a complete loan approval prediction system using machine learning, from data preprocessing to deployment. The current model provides useful predictive performance and a user-friendly interface suitable for academic demonstration and foundational real-world adaptation.

Prepared for: BCA Final Year Project Documentation

Date: 2026-05-04